

IndicMOS Evaluation Report

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Overview

This report summarizes the evaluation of IndicMOS against human Mean Opinion Score (MOS) ratings across English, Hindi, Kannada, and Telugu. The evaluation compares predicted and human scores using correlation and error metrics. Detailed methodology, implementation, and experiment setup are documented in the project README.

Results Summary

Language	n	Pearson r	Spearman ρ	Kendall τ	MAE	RMSE	Listeners
English	6	0.724	0.696	0.552	0.599	0.683	27
Hindi	6	-0.113	-0.086	-0.067	0.820	1.063	18
Kannada	6	0.723	0.794	0.645	0.486	0.573	7
Telugu	6	0.278	0.143	0.067	0.874	0.919	11
Pooled	24	0.333	0.423	0.272	0.695	0.832	39

Key Findings & Interpretation

- **Kannada** showed the strongest agreement between IndicMOS predictions and human listener ratings. It achieved the highest rank correlations (Spearman = 0.794, Kendall = 0.645) along with the lowest MAE (0.486) and RMSE (0.573). This indicates that IndicMOS was able to rank audio samples similarly to human listeners while also producing prediction values that were relatively close to the actual MOS scores. Although the correlations narrowly missed the conventional significance threshold ($p < 0.05$), the results suggest a strong positive trend that would likely become statistically significant with a larger evaluation dataset.

- **English** demonstrated a moderately strong positive relationship between IndicMOS and human ratings (Pearson = 0.724, Spearman = 0.696). This suggests that higher-quality speech, as perceived by listeners, generally received higher IndicMOS scores as well. The prediction errors (MAE = 0.599, RMSE = 0.683) were reasonably low, indicating acceptable prediction accuracy. However, due to the limited sample size (6 audio samples), these correlations were not statistically significant, so the findings should be considered encouraging rather than conclusive.

- **Hindi** showed the weakest performance among all evaluated languages. Correlation values were close to zero (and slightly negative), indicating that IndicMOS failed to preserve the same quality ranking as human listeners. Additionally, Hindi recorded the highest prediction errors (MAE = 0.820, RMSE = 1.063), suggesting that the predicted MOS values deviated substantially from the human ratings. These results imply that, under the current evaluation setup, IndicMOS is not a reliable predictor of perceived speech quality for Hindi.

- **Telugu** exhibited only a weak positive relationship between IndicMOS predictions and human judgments. Both the correlation coefficients and error metrics indicate that the model struggled to consistently rank speech samples according to listener preferences. While the performance was better than Hindi, the relatively high MAE (0.874) and RMSE (0.919) suggest that IndicMOS still has limited predictive accuracy for Telugu speech.
- **Overall (Pooled Analysis):** Combining data from all four languages resulted in a statistically significant Spearman rank correlation ($\rho = 0.423$, $p = 0.0397$), indicating that IndicMOS captures some overall ranking trends across the complete dataset. However, this pooled result should be interpreted with caution because it masks substantial language-wise variation. The strong performance in Kannada and moderate performance in English partially offset the weak performance in Hindi and Telugu. Therefore, the per-language analysis provides a more accurate assessment of IndicMOS than the pooled statistics alone.

Conclusion

The evaluation suggests that **IndicMOS exhibits language-dependent performance** rather than consistent behavior across all Indian languages. It appears most reliable for Kannada and reasonably promising for English, while its predictions for Hindi and Telugu show weaker alignment with human perception. Since each language was evaluated using only six audio samples, the statistical power of the analysis is limited. A larger and more diverse evaluation dataset would enable more robust conclusions regarding the reliability and generalizability of IndicMOS across different languages.

Project Repository

GitHub Repository: <https://github.com/arcyadav/IndicMOS-evaluation.git>